



SALESSENSE-AI



PROBLEM STATEMENT

Business Problem

Retail businesses often struggle to answer:

- How much product will be sold next month?
- Which products should be stocked more?
- Are there unusual sales patterns?
- How can inventory costs be reduced?

Objective

Develop an AI-powered system that:

- Forecasts future sales
- Detects anomalies
- Segments product demand
- Provides an interactive business dashboard



EXPLORATORY DATA ANALYSIS

Key Insights

- Four years of retail sales analyzed
- Strong seasonal sales trends observed
- Technology category generated the highest revenue
- Monthly sales showed consistent growth
- Regional sales compared for performance

Visualizations

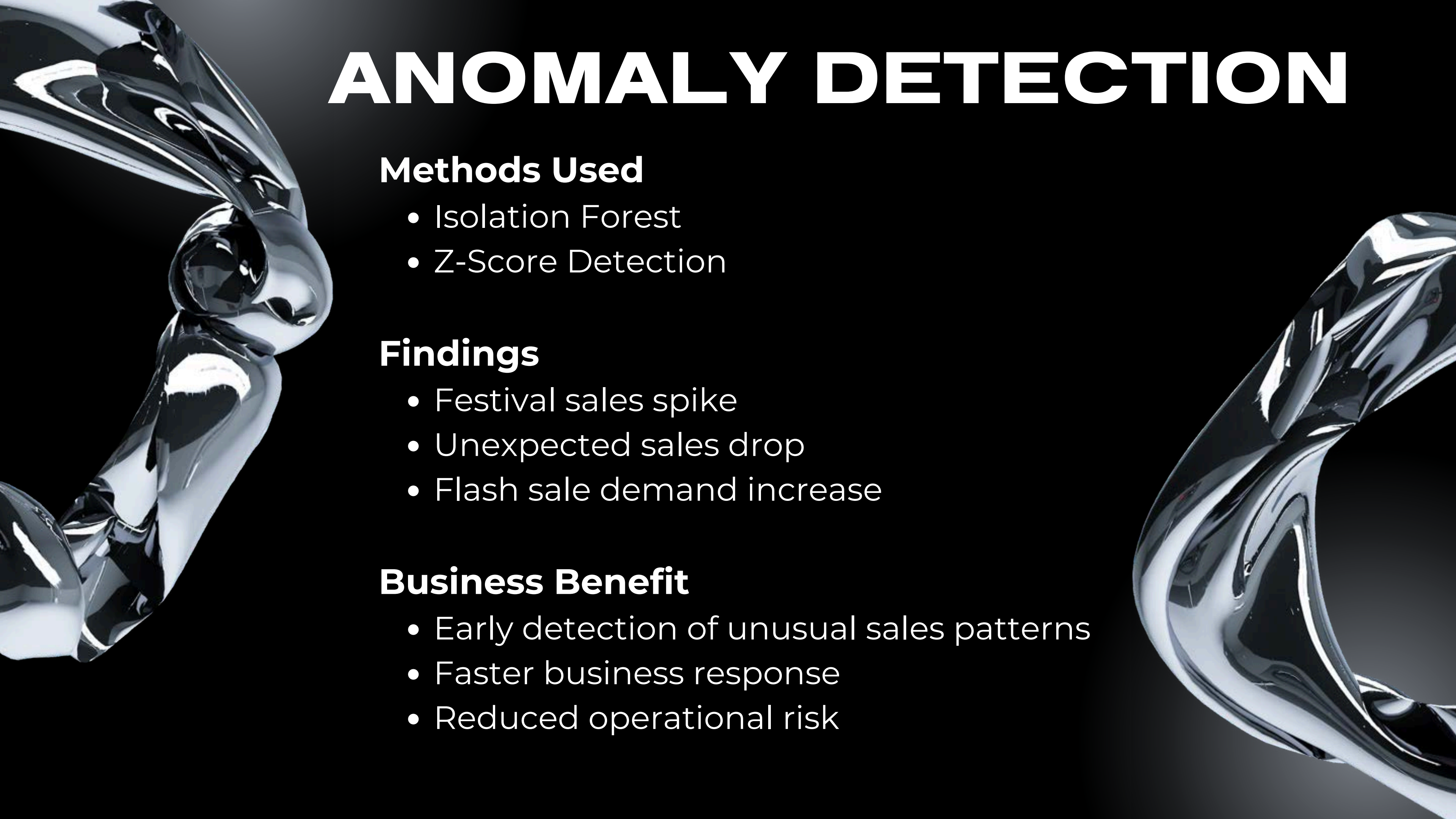
- Sales by Year
- Monthly Sales Trend
- Sales by Region
- Category Analysis



FORECASTING MODELS

Models Compared

- **SARIMA** : Statistical forecasting model
- **Prophet** : Facebook time-series forecasting
- **XGBoost** : Machine Learning forecasting using lag features
- **Evaluation Metric** : MAE, RMSE
- **Best Model**: XGBoost

The background of the slide features abstract, flowing blue liquid splashes on a black background, creating a dynamic and modern aesthetic. The liquid appears to be moving from the top left and bottom right towards the center, with highlights and reflections giving it a three-dimensional feel.

ANOMALY DETECTION

Methods Used

- Isolation Forest
- Z-Score Detection

Findings

- Festival sales spike
- Unexpected sales drop
- Flash sale demand increase

Business Benefit

- Early detection of unusual sales patterns
- Faster business response
- Reduced operational risk



DEMAND SEGMENTATION

Product Clusters

High Volume – Stable Demand

- Maintain high inventory.

Growing Demand

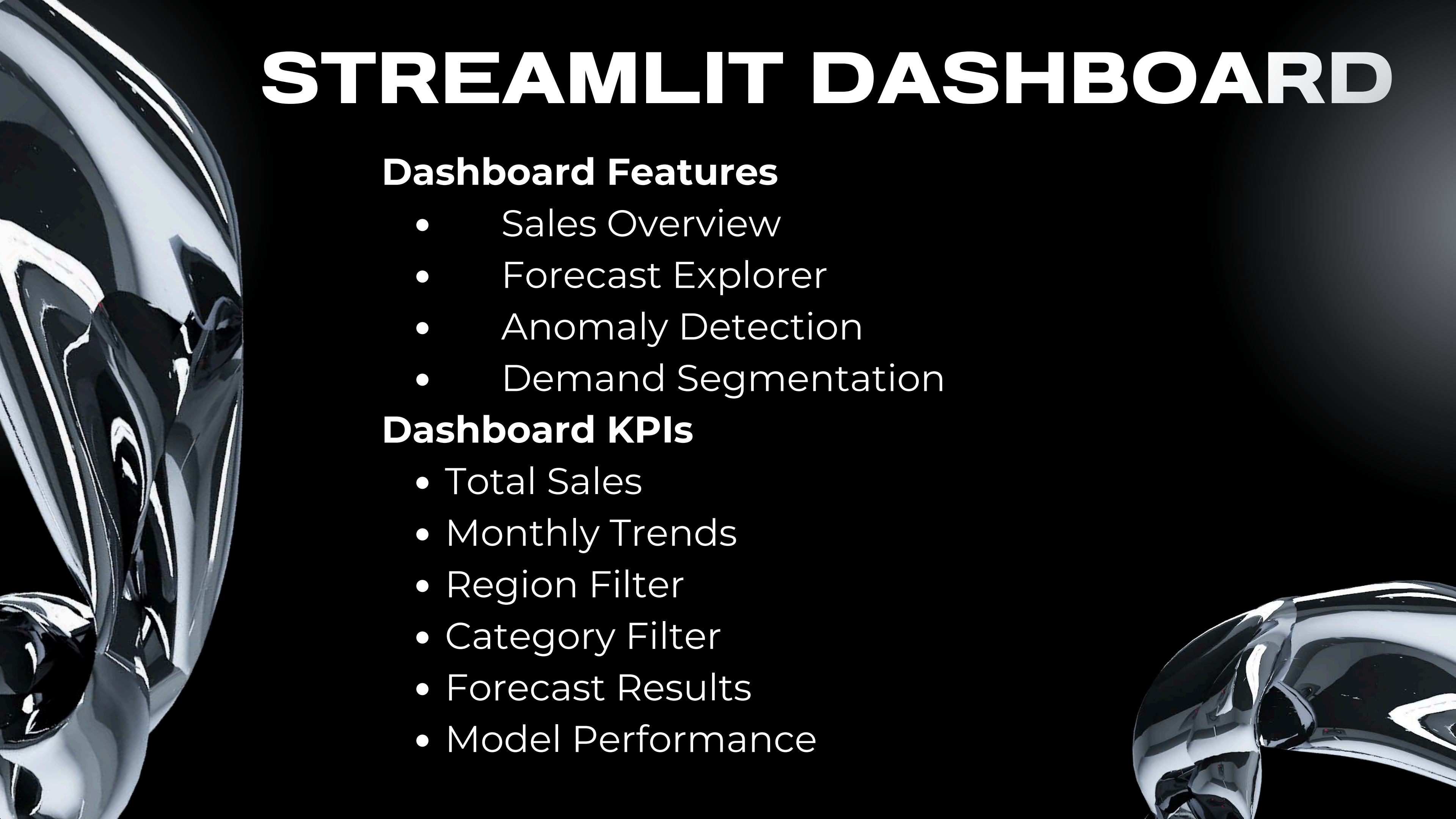
- Increase stock gradually.

Low Volume – High Volatility

- Use demand-based replenishment.

Declining Demand

- Reduce inventory and promote clearance sales.
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STREAMLIT DASHBOARD

Dashboard Features

- Sales Overview
- Forecast Explorer
- Anomaly Detection
- Demand Segmentation

Dashboard KPIs

- Total Sales
- Monthly Trends
- Region Filter
- Category Filter
- Forecast Results
- Model Performance

MACHINE LEARNING MODELS

The following models were trained and evaluated:

- Linear Regression
- Random Forest Regressor
- XGBoost Regressor
- Prophet (if used)
- Time Series Models (if applicable)

Best Model

- XGBoost

CONCLUSION

Project Outcomes

Built an end-to-end forecasting system

Compared three forecasting models

Selected XGBoost as the best model

Detected sales anomalies

Segmented products using Machine Learning

Developed an interactive Streamlit dashboard

FUTURE ENHANCEMENTS

- Real-time sales forecasting
- Automated monthly model retraining
- Integration with ERP and inventory management systems
- AI-powered demand recommendation engine
- Live business intelligence dashboard with streaming data





THANK YOU